Deepfake Audio Detector

Machine Learning Model Comparison and Evaluation

Progress Report 3 | October 2026

KEY FINDING

MLP delivered the strongest overall results, reaching 99.1% accuracy with only 3 false negatives in the 20,000-sample test.

1. Project Overview

Generative AI has become increasingly capable of producing realistic fake audio. Because these recordings can be difficult for people to identify by listening alone, they create risks related to misinformation, impersonation, and online deception. Our project is developing a deepfake audio detector that uses machine learning to classify audio as real or spoofed.

2. Experimental Design

Five machine learning methods were evaluated: Random Forest, Support Vector Machine (SVM), Gradient Boosting, Logistic Regression, and Multi-Layer Perceptron (MLP). Each model was tested at three sample sizes - 1,000, 10,000, and 20,000 samples - so performance could be compared as the amount of training data increased. The experiments were conducted in Google Colab.

Models Evaluated

Random Forest - combines many decision trees and uses their combined vote to classify the audio.
Support Vector Machine (SVM) - finds a boundary that best separates real and spoofed samples.
Gradient Boosting - builds models in sequence, with each new model correcting earlier errors.
Logistic Regression - estimates the probability that a sample belongs to the real or spoofed class.
Multi-Layer Perceptron (MLP) - uses a neural network to learn more complex patterns in the input data.

False Positive	False Negative
A real audio file incorrectly labeled as spoofed.	A spoofed audio file incorrectly labeled as real.

Why false negatives matter: In a deepfake detector, a false negative allows fake audio to pass as authentic, so this error type is especially important when comparing models.

3. Training Results

The tables below summarize runtime, accuracy, false positives, and false negatives for each model at the three tested sample sizes.

1,000 Samples

Model	Runtime	Accuracy	False Positives	False Negatives
Random Forest	2 min	97.2%	2	3
SVM	2 min	98.8%	1	1
Gradient Boosting	1 min	96.6%	3	3
Logistic Regression	1 min	96.08%	4	3
MLP	**2 min**	**99.4%**	**0**	**1**

Result: MLP produced the highest accuracy (99.4%) with 0 false positives and 1 false negative. SVM was the next strongest model at 98.8%.

10,000 Samples

Model	Runtime	Accuracy	False Positives	False Negatives
Random Forest	5 min	98.3%	2	17
SVM	5 min	99.3%	4	4
Gradient Boosting	4 min	98.9%	4	8
Logistic Regression	4 min	96.6%	20	19
MLP	**5 min**	**99.3%**	**3**	**5**

Result: MLP and SVM tied at 99.3% accuracy. SVM had 4 false negatives, while MLP had 5; MLP had slightly fewer false positives.

20,000 Samples

Model	Runtime	Accuracy	False Positives	False Negatives
Random Forest	8 min	97.7%	8	29
SVM	9 min	98.6%	9	13
Gradient Boosting	7 min	98.3%	11	15
Logistic Regression	7 min	95.2%	27	49
MLP	**8 min**	**99.1%**	**10**	**3**

Result: MLP returned the highest accuracy (99.1%) and the lowest number of false negatives (3). Logistic Regression produced the lowest accuracy and the most classification errors.

4. Performance Comparison

Across all three sample sizes, MLP remained at or near the top of the comparison. It achieved the highest accuracy at 1,000 samples, tied SVM for the highest accuracy at 10,000 samples, and again ranked first at 20,000 samples.

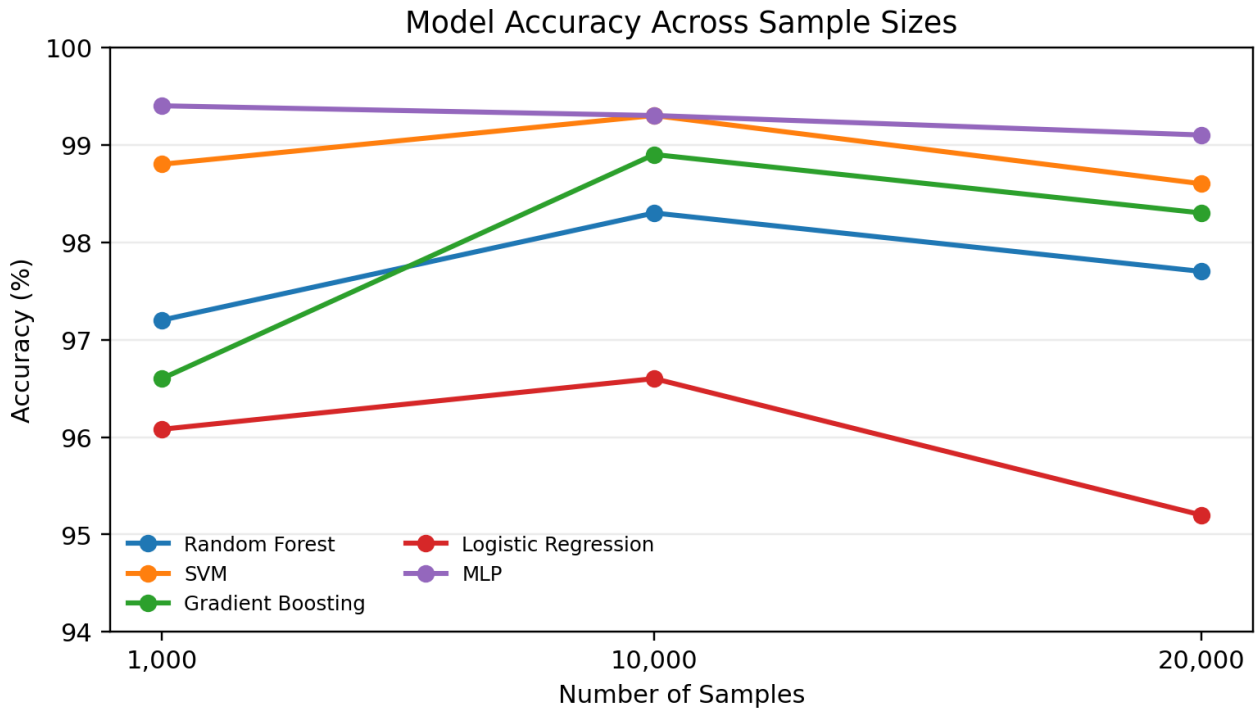


Figure 1. Accuracy of all five models across the three sample sizes.

5. Model Analysis

Model	1K Accuracy	20K Accuracy	20K False Negatives
Random Forest	97.2%	97.7%	29
SVM	98.8%	98.6%	13
Gradient Boosting	96.6%	98.3%	15
Logistic Regression	96.08%	95.2%	49
MLP	**99.4%**	**99.1%**	**3**

MLP provided the strongest overall balance of accuracy and error control. Its accuracy stayed above 99% at every sample size, and in the largest test it produced only 3 false negatives. This is particularly important because false negatives represent spoofed audio being accepted as real.

SVM also performed consistently well and remained close to MLP throughout the experiments. Gradient Boosting produced competitive results, while Random Forest showed a larger increase in false negatives at 20,000 samples. Logistic Regression was the weakest performer overall, with the lowest 20,000-sample accuracy and the highest number of false negatives.

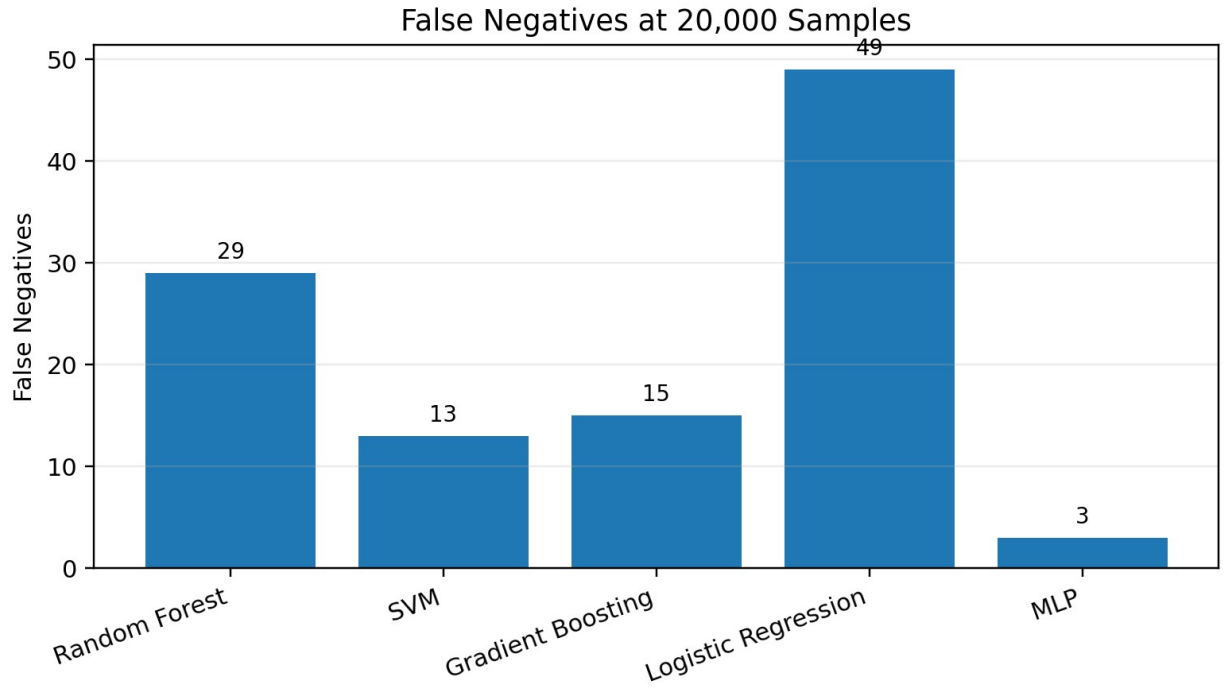


Figure 2. False negatives in the 20,000-sample test. Lower is better.

6. Current Progress and Next Steps

Completed	Five machine learning models compared at 1,000, 10,000, and 20,000 samples.
Current Result	MLP is the strongest candidate based on overall accuracy and false-negative performance.
Next Steps	Use the selected model in the detector, continue integration with the application, and perform end-to-end testing with uploaded audio.

7. Progress Report Conclusion

The model-comparison stage is complete and provides a clear direction for the next phase of the project. MLP achieved the highest or tied-highest accuracy at every tested sample size and performed especially well at 20,000 samples, where it recorded only 3 false negatives. Based on these results, MLP is the leading model for continued development of the deepfake audio detector.